

Connor Day

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EDUCATION

University of California, San Diego

San Diego, CA

Bachelor of Science in Data Science

Expected June 2028

- **Relevant Coursework:** Probability, Theoretical Foundations of Data Science I & II, Exploratory Data Analysis and Inference, Systems for Scalable Analytics, Data Visualization, Data Management (SQL), Practice and Application of Data Science

PROJECTS

Garment Search | *Python, PyTorch, Hugging Face Transformers, FAISS, Pandas, NumPy* Sep. 2026 – Oct. 2026

- Built a text-to-image search engine over 45,622 Fashionpedia clothing photos, comparing fashion-specific FashionCLIP against general-purpose CLIP using exact FAISS cosine-similarity indexes
- Pre-registered the evaluation before embedding any images: 20 queries across four difficulty tiers, written relevance criteria, pass thresholds, and six predicted failure modes
- FashionCLIP outperformed CLIP on capped recall@10 (0.612 vs. 0.488) and hand-judged precision@10 (0.755 vs. 0.540), but both failed the compositional tier and neither returned a clearly oversized blazer (0 of 10)
- Hand-judged 391 query-image pairs blind to model and rank, finding that 28% of results the dataset's labels scored as irrelevant were clearly relevant, so label-based recall understated retrieval quality

Thyroid and Metabolic Health | *Python, Pandas, Polars, statsmodels, scikit-learn, Matplotlib* Aug. 2026

- Pre-registered four falsifiable hypotheses linking thyroid function to metabolic health, then tested them on three pooled NHANES cycles (2007–2012) covering 3,459 U.S. adults
- Fit survey-weighted regression models using NHANES strata, PSUs, and sample weights, adjusting for age, sex, and BMI, and built a metabolic syndrome flag from the 2009 Harmonized clinical criteria
- Applied FDR and Bonferroni correction: three of four hypotheses were unsupported, and the one significant result ran opposite to the prediction (+14.9 mg/dL triglycerides per unit FT3/FT4 ratio, $p < 0.001$)
- Built a reproducible pipeline from raw NHANES survey files to per-hypothesis analytic samples

Power Outages and Population Density | *Python, Pandas, scikit-learn, Plotly* Course Project, June 2026

- Tested whether population density predicts outage duration across 1,534 major U.S. outages (2000–2016); a 10,000-iteration permutation test found no significant difference ($p = 0.065$)
- Built a scikit-learn Random Forest pipeline tuned with 5-fold GridSearchCV on features known at outage start, cutting test RMSE 12% below a mean-only baseline (6,888 vs. 7,839 minutes)
- Used permutation tests to analyze missingness in customer counts and check model fairness across density groups

EXPERIENCE

Polo Ralph Lauren Factory Store

Livermore, CA

Front & Back of House Team Member

June 2022 – Sep. 2025

- Presented floor sets and store walks to store management and visiting corporate leadership, explaining the reasoning behind layout and merchandising decisions
- Selected by management to collaborate with top-performing team members on styling store sections for corporate lookbook photography, adopted company-wide as the seasonal floor standard for all Ralph Lauren factory stores
- Trained 10+ new team members on store procedures, POS operations, and customer service standards, explaining processes clearly to people with no prior retail experience

TECHNICAL SKILLS

Languages: Python, SQL (proficient); Java, R, C++ (coursework)

Libraries: Pandas, NumPy, Polars, statsmodels, scikit-learn, PyTorch, Hugging Face, FAISS, Plotly, Matplotlib

Data Skills: Hypothesis testing, survey-weighted regression, multiple-comparison correction, retrieval evaluation

Tools: Git, GitHub, Jupyter, LaTeX, Excel, PowerPoint